Neurobiology BIOL 4200-001 **FALL 2021**
Neurobiology BIOL 4200-002 (HON)
Neurobiology BIOL 6200-001 August 18 – December 7

BIOL 4200/4200H/6200: Neurobiology, 3 credits

- Class Meetings:
  - All sections: MW, 3:35-4:45pm, Freeman Hall, Room 138
  - FOR ZOOM MEETINGS: Access to Zoom is on the left of Canvas.
- Teaching Modality:
  - This course will be taught in the traditional modality.
  - Class will meet in-person at the scheduled time.

INSTRUCTOR INFORMATION

Julia George, PhD
Office: 231 Long Hall
Email: jgeorg4@clemson.edu
Phone: 864-656-2328 (Biological Sciences main office)
OFFICE HOURS: virtual via Zoom, Thursdays from 2:00-3:00pm, or by appointment

COURSE DESCRIPTION

From the student handbook: (provides a) "Broad background in neurobiology. Topics include neuroanatomical structure-function; conduction in the neuron; neurite growth and development; neuromuscular junction; chemistry, physiology, and pharmacology of specific neurotransmitters and receptors; visual process; axoplasmic transport; hypothalamic-pituitary regulation; theories of behavior; theories of learning and memory."

LEARNING OUTCOMES

By the end of this course, students will:

1. Understand the fundamental electrical properties of neurons and glia, and the mechanisms of chemical and electrical signaling by cells in the nervous system.
2. Understand the molecular basis of synaptic plasticity, which underlies learning and memory.
3. Understand fundamental mechanisms of sensation.
4. Understand basic mechanisms of neural development, repair and regeneration.
5. Understand to basic mechanisms of behavioral regulation.

REQUIRED KNOWLEDGE

Computer skills: Web browser, Canvas, Google Apps, Zoom
Prerequisite: BIOL 1040 and BIOL 1060; or BIOL 1110.
Prerequisite or concurrent enrollment: BIOL 4610.

REQUIRED TEXTS and RESOURCES:

- **TEXTBOOK**: From Neuron to Brain, 6th edition, 2021. A. Robert Martin, David A. Brown, Mathew E. Diamond, Antonino Cattaneo, Francisco F. De-Miguel. Oxford University Press, USA. ISBN-13: 978-0197542774; ISBN-10: 0197542778
- **DIGITAL MATERIALS**: you will need access to the digital materials that accompany the assigned textbook. RECOMMENDED: A redemption code is included with e-book rental through the campus bookstore. ALTERNATIVE: If you decide to purchase a hardcover book, it will include a code that can only be redeemed once for access to digital materials. You should anticipate that a used book may not have a useable code.

TOPICAL OUTLINE

This is a preliminary schedule that is subject to change. See Canvas for reading and homework assignments.

Date	Day	Topic
August 18	Wed	Introduction to course
August 23	Mon	Ion channels and signaling
August 25	Wed	Structure of ion channels
August 30	Mon	Ionic basis of the resting potential
September 1	Wed	Ionic basis of the action potential
September 6	Mon	Electrical signaling in neurons
September 8	Wed	Ion transport across cell membranes
September 13	Mon	Properties and functions of neuroglial cells
September 15	Wed	Exam 1
September 20	Mon	Mechanisms of direct synaptic transmission
September 22	Wed	Indirect mechanisms of synaptic transmission
September 27	Mon	Release of neurotransmitters at synapses
September 29	Wed	Neurotransmitters in the central nervous system
October 4	Mon	Transmitter synthesis, storage, transport, and inactivation
October 6	Wed	Synaptic plasticity
October 11	*Mon*	*Fall break, no class*
October 13	Wed	The molecular and cellular biology of synaptic plasticity
October 18	Mon	Exam 2
October 20	Wed	Sensory transduction

October 25	Mon	Transduction and transmission in the retina
October 27	Wed	Touch, pain, and texture sensation
November 1	Mon	Auditory and vestibular sensation
November 3	Wed	Constructing sensation
November 8	Mon	Initiation and control of coordinated muscular movements
November 10	Wed	Exam 3
November 15	Mon	Development of the nervous system
November 17	Wed	Critical periods in sensory systems
November 22	Mon	Regeneration and Repair of synaptic connections after injury
November 24	*Wed*	*Thanksgiving break, no class*
November 29	Mon	Special Topics
December 1	Wed	Special Topics
December 7	Tues	Final exam, 7-9:30pm

COURSE POLICIES

Grading:

Grading Scale:

A	90-100%
B	80-89%
C	70-79%
D	60-69%
F	0-59%

Grading Activities: There will be 4 exams that account for 80% of your grade (3 hourly exams and a comprehensive final exam, each worth 20%). 20% of your grade will be based on homework/in-class assignments. Any student having an A in all classwork at the end of the semester is exempt from the final exam.

Honors students: Honors students will complete an additional project, which will substitute for half of the homework grade (10% of final grade). Students will choose an article related to neurobiology from the news or popular press, then write a critical evaluation of the scientific evidence underlying the news article. The paper should be 3-5 pages (excluding references), single-spaced, half-inch margins, 10 pt Arial font, with no spaces between paragraphs. The following will be weighted equally for evaluation: adherence to format, social and cultural implications of the topic, presentation of the news article, presentation of the relevant research, critical comparison of the two.

Graduate students: Graduate students will complete an additional project, which will substitute for half of the homework grade (10% of final grade). Each student will select a current journal article on a topic in neurobiology, to be approved by the instructor, and

prepare a 25-minute presentation to be presented in class during the last week of class. The following will be weighted equally for evaluation: background explaining the importance of the topic and current knowledge, description of the major hypotheses tested, explanation of the most important methodologies employed, results including explanation of major figures, and critical evaluation of the conclusions, important future experiments, and ethical considerations.

Attendance:

You must participate during the first week of class. If you have not participated by AUGUST 24 (last day to add a class) or if you accumulate excessive absences (more than 3 unexcused absences), you may be dropped from the course.

You should speak with me as soon as possible regarding any absence and develop a plan for any make-up work. It is your responsibility to let me know if you need to make any changes due to the pandemic. In the event of an emergency, you should make direct contact with me, preferably before a class or an exam takes place. You may find it helpful to use the Notification of Absence module in Canvas. If you have difficulty or are unable to electronically report the absence, you may call the Office of Advocacy and Success at 864-656-0935 for assistance and guidance.

If I need to miss a class due to illness or some other unavoidable circumstance, I will make every effort to notify you in advance. However, should I be absent for a scheduled class, you should wait for 15 minutes before assuming that class is cancelled.

Inclement Weather:

Any exam that was scheduled at the time of a class cancellation due to inclement weather, University power outage, etc. will be given at the next class meeting unless contacted by the instructor. Any assignments due at the time of a class cancellation due to inclement weather will be due at the next class meeting unless the instructor contacts students. Any extension or postponement of assignments or exams must be granted by the instructor via email or Canvas within 24 hours of the weather-related cancellation.

Academic Continuity:

Clemson has developed an Academic Continuity Plan for academic operations. Should university administration officially determine that the physical classroom facility is not available to conduct classes, class will be conducted in a virtual (online) form. The university issues official disruption notifications through email, website, text notification and Social Media. When notified, use one of the following links to navigate to Clemson Canvas where you will find important information about how we will conduct class:

- Primary access link: http://www.clemson.edu/canvas
- Secondary access link, if needed: https://clemson.instructure.com/
- You can also use the Canvas Student App. Visit the downloads page for this app.

Course activities will occur through the Canvas course.

Expectations:

COVID-19:

To maintain an inclusive and safe environment so that all members of this course may succeed, it is also expected that those in attendance follow health guidelines for everyone's safety. As biologists that are working to understand concepts of ecology, immunology, and disease spread, all students should wear masks while inside the classroom. There is currently a mandate to wear masks inside university buildings for the first 3 weeks of the semester. In the absence of a mandate (should the mandate expire or be reversed) the use of masks will still be expected. It is a sign of understanding biological principles, public health responsibilities, and looking out for your fellow students and their families that may not be able to be vaccinated yet.

Should a student feel ill or have received a positive COVID test, that student should not attend class. Instead, notify your instructor that you are ill via the Notification of Absence module in Canvas. Students should follow up with their instructor to develop a continued plan of study if their illness impacts their ability to participate in regular course work. While in-person class attendance is best, the instructor has posted materials on Canvas for use by those students who cannot attend due to illness. Keep in mind that students will be required to provide documentation for any absences to be approved, especially those impacting exam days (see the grade section above for more details).

If the instructor is required to isolate due to Covid, you will be notified of alternative arrangements via Canvas. Depending on the length of the anticipated absence, this could include a substitute in-person instructor, cancellation of a scheduled lecture, assignment of an alternate activity, or one or more online (Zoom) lectures.

As we continue to learn more about the state of the pandemic and the current level of risk, policies to mitigate the spread of disease in our classroom may change throughout the semester. Your instructor will inform you of these changes ahead of time so that you can prepare accordingly.

Please be familiar with the information on the Healthy Clemson website.

Conduct:

Please use appropriate conduct to help maintain a safe learning environment. Participants should:

- *Never transmit or promote content known to be illegal.*
- *Respect people's privacy as well as your own.*
- *Forgive other people's mistakes.*
- *Never use harassing, threatening, embarrassing, or abusive language or actions.*

UNIVERSITY POLICIES

Accessibility Services:

Clemson University values the diversity of our student body as a strength and a critical component of our dynamic community. Students with disabilities or temporary injuries/conditions may require accommodations due to barriers in the structure of facilities, course design, technology used for curricular purposes, or other campus resources. Students who experience a barrier to full access to this class should let the instructor know and make an appointment to meet with a staff member in Student Accessibility Services as soon as possible. You can make an appointment by calling 864-656-6848, by emailing studentaccess@lists.clemson.edu, or by visiting Suite 239 in the Academic Success Center building. Appointments are strongly encouraged – drop-ins will be seen if at all possible, but there could be a significant wait due to scheduled appointments. Students who have accommodations are strongly encouraged to request, obtain and send these to their instructors through their AIM portal as early in the semester as possible so that accommodations can be made in a timely manner. It is the student's responsibility to follow this process each semester.

You can access further information at the Student Accessibility website. Other information is at the university's Accessibility Portal.

Non-Discrimination:

The Clemson University Title IX statement: Clemson University is committed to a policy of equal opportunity for all persons and does not discriminate on the basis of race, color, religion, sex, sexual orientation, gender, pregnancy, national origin, age, disability, veteran's status, genetic information or protected activity in employment, educational programs and activities, admissions and financial aid. This includes a prohibition against sexual harassment and sexual violence as mandated by Title IX of the Education Amendments of 1972. This Title IX policy is located on the Campus Life website. Ms. Alesia Smith is the Clemson University Title IX Coordinator, and the Executive Director of Equity Compliance. Her office is located at 223 Brackett Hall, 864.656.0620. Remember, email is not a fully secured method of communication and should not be used to discuss Title IX issues.

Academic Integrity:

As members of the Clemson University community, we have inherited Thomas Green Clemson's vision of this institution as a "high seminary of learning." Fundamental to this vision is a mutual commitment to truthfulness, honor, and responsibility, without which we cannot earn the trust and respect of others. Furthermore, we recognize that academic dishonesty detracts from the value of a Clemson degree. Therefore, we shall not tolerate lying, cheating, or stealing in any form.

All infractions of academic dishonesty by undergraduates must be reported to Undergraduate Studies for resolution through that office. In cases of plagiarism instructors may use the Plagiarism Resolution Form.

See the Undergraduate Academic Integrity Policy website for additional information and the current catalogue for the policy.

For graduate students, see the current graduate student handbook for all policies.

Copyright Statement:

Materials in some of the courses are copyrighted. They are intended for use only by students registered and enrolled in a particular course and only for instructional activities associated with and for the duration of the course.

Student Safety:

Clemson University is committed to providing a safe campus environment for students, faculty, staff, and visitors. As members of the community, we encourage you to take the following actions to be better prepared in case of an emergency:

a. Ensure you are signed up for emergency alerts (https://www.getrave.com/login/clemson),
b. Download the Rave Guardian app to your phone (https://www.clemson.edu/cusafety/cupd/rave-guardian/)
c. Learn what you can do to prepare yourself in the event of an active threat(http://www.clemson.edu/cusafety/EmergencyManagement/)

Helpful Resources:

The Department of Biological Sciences is committed to providing a supportive learning environment for all students. If you are facing tough times, please utilize these resources. You may also reach out to me directly.

a. Financial Assistance: Clemson Student Financial Aid (http://www.clemson.edu/financial-aid/index.html), and Anderson Interfaith Ministries (https://www.aimcharity.org/)
b. Food Insecurity: Clemson Paw Pantry (http://facebook.com/CUpawpantry)
c. Textbook Assistance: Clemson Library FAQs (https://clemson.libanswers.com/faq/100017) and Clemson Bookstore Open Education Resources copies (https://clemson.bncollege.com/shop/clemson/page/find-oer)
d. General Resources: (https://www.clemson.edu/studentaffairs/advocacy-success/resources/index.html)
e. Multicultural Resources and Support: Gantt Multicultural Center (https://www.clemson.edu/centers-institutes/gantt/multicultural-programs)
f. LGBTQ Support: Gantt Multicultural Center –LGBTQ Programs (https://www.clemson.edu/centers-institutes/gantt/lgbtq-programs/resources.html)
g. Adjustment and Transition: Counseling and Psychological Services (https://www.clemson.edu/campus-life/student-health/caps) and Student Transitions and Family Programs (http://www.clemson.edu/studentaffairs/stfp/index.html)
h. Interpersonal Violence: Healthy Campus (https://www.clemson.edu/campus-

life/healthy-campus/interpersonal-violence/index.html)

i. Addiction and Recovery: SC: (https://www.daodas.sc.gov/), Web: (https://addictionresource.com/) and Vaping (https://vapingdaily.com/health/).